

aegis.

Trust the person. Not just the pixels.

Problem. A realistic face on a screen is no longer reliable evidence of a live person. Entrust reported a 40% year-over-year increase in digital injection attacks. Pindrop's September 2026 survey found that 74% of surveyed US enterprise security leaders had encountered or suspected a deepfake attack in the previous year. Gartner predicted in 2024 that, by 2026, 30% of enterprises would consider identity-verification and authentication solutions unreliable in isolation. These are differently scoped external findings, not measurements of Aegis. [1-3]

Solution. Aegis is a proposed layered presence-verification service with an explainable decision: approve, step-up, or reject. This browser-only prototype combines camera metadata, randomized head/blink prompts, a partial mouth-motion challenge, and an optional screen-light response. It makes every signal's origin visible and treats missing evidence as uncertainty. Deepfake-artifact and audio-visual model outputs are clearly simulated, as are the attack examples and administrator history. No biometric upload, microphone capture, recording, or persistent session storage is implemented.

Why layers can beat a standalone check. A face match asks whether an image resembles someone; it does not by itself establish who supplied the stream or whether a live person followed a fresh challenge. Complementary checks can address different failure modes and support targeted step-up instead of a brittle yes/no answer. But adding scores does not automatically create independent evidence: Aegis's camera-derived checks share a source and remain spoofable. The full product must validate incremental benefit against adaptive attacks, across devices and user groups, rather than multiply unproven confidence claims.

First target users. The initial customer hypothesis is security and fraud teams responsible for high-risk account recovery, fintech onboarding, and enterprise help-desk impersonation. Aegis would complement their identity stack, not replace document verification, credential authentication, or human review. Early pilots would focus on one workflow with a clear escalation path and consenting participants.

Business model. A business-to-business subscription with a platform fee and usage-based verification pricing, followed by enterprise SDK, workflow, and support tiers. This is a proposed model, not existing revenue, contracted customers, or validated willingness to pay. Pilot success would be measured through false accept/reject tradeoffs, accessible completion, reviewer workload, latency, and validated attack resistance, not a headline accuracy number.

What the hackathon proves. An end-to-end interaction, local landmark-based heuristics, explicit simulation boundaries, deterministic fixture decisions, and an inspectable risk policy. Live sessions deliberately cannot approve because key evidence is missing. The next build adds server-owned sessions and policy, a license-reviewed real model, stronger and more accessible liveness, and a documented API/SDK. The browser alone cannot become a trusted authorization boundary.

The ask. A design partner with one recovery or onboarding workflow, consenting test participants, and a security reviewer to help define a credible evaluation. Aegis aims to make uncertainty visible before granting trust.

[1] Entrust, 2026 Identity Fraud Report announcement, November 18, 2025.

[2] Pindrop, enterprise survey announcement, September 28, 2026; 250 US security leaders.

[3] Gartner, February 1, 2024 forecast; a prediction, not an observed 2026 result.